

Quantifying the Performance Cost of Replacing Registry Metadata with Street-View-Derived Building Attributes in Archetype-Based UBEM Workflows

Scientific work to obtain the degree

Master of Science (M.Sc.)

at the TUM School of Engineering and Design
of the Technical University of Munich.

Supervised by	Agata Dalach Jose Quesada Allerhand Chair of Computing in Civil and Building Engineering
Submitted by	Pei-Ling Song (03798081) Franz-Kötterl-Straße 19a 80995 München e-Mail: peiling.song@gmail.com
Submitted on	August 22, 2026

Abstract

Work in progress. The current Draft V8 does not yet contain an abstract.

Contents

Abstract	I
1 Introduction	1
1.1 Background and Motivation	1
1.2 Research Objectives	2
2 Literature Review	4
2.1 Comparison Framework	4
2.2 Reference and Open-Data Approaches	6
2.3 Proxy-Based Attribute and Archetype Inference	6
2.4 Direct Image-to-EPC Approaches	7
2.5 Image-Based Building-Attribute Extraction	7
2.5.1 Supervised Full Fine-Tuning	8
2.5.2 Frozen Self-Supervised Representations	8
2.5.3 VLM Inference Without Parameter Updating	8
2.6 Archetype-Grounded and Semi-Synthetic Datasets	9
3 Method	10
3.1 Overview	10
3.1.1 Pipeline Architecture	10
3.1.2 Semi-Synthetic Dataset: Three-Layer Structure	11
3.2 Dataset Construction	12
3.2.1 Sources	12
3.2.2 SVI Acquisition, Data Linkage, and Screening Rules	13
3.2.3 Data Partitioning and Cross-Validation	17
3.3 Vision Inference Strategies and Training Configurations	17
3.4 TABULA Archetype Matching	20
3.4.1 NL Matrix Structure	20
3.4.2 Attribute-to-Archetype Mapping	22
3.4.3 U-Value Lookup and As-Built Assumption	23
3.5 Downstream EPC Classification Model	23
4 Experiments and Evaluation	25
4.1 Strategy and Metrics	25
4.1.1 Dataset Statistics	25
4.1.2 Route Families and Evaluation Settings	26
4.1.3 Prediction Tasks and Metrics (WIP)	28
4.1.4 Physical-Consequence Metric (WIP)	28
4.2 Quantitative Results	28
4.2.1 Upstream Attribute Extraction	28
4.2.2 Joint-Cell Assignment	29

4.2.3	Downstream Route Comparison	30
4.2.4	Physical-Consequence Scoring (WIP)	32
4.2.5	Amsterdam-Held-Out Transfer Evaluation (WIP)	33
4.3	Qualitative and Diagnostic Analysis (WIP)	33
4.3.1	Error Mechanism: Regression to the Mean in Construction Year (WIP)	33
4.3.2	Feature Ablation and Marginal Utility (WIP)	34
4.3.3	Extractability–Utility Audit (WIP)	34
4.3.4	Label-Quality Audit (WIP)	35
4.3.5	Pool-Composition Audit (WIP)	35
5	Discussion and Conclusion (WIP)	36
5.1	Observations (WIP)	36
5.1.1	Seven-Class Ceiling: Task Structure Rather Than Visual Failure (WIP)	36
5.1.2	Unit-of-Observation Mismatch in EPC Labels (WIP)	36
5.1.3	Boundary of the Role of SVI: Division of Labour Between Modalities (WIP)	36
5.2	Limitations (WIP)	37
5.3	Future Work (WIP)	37
5.4	Conclusion	37
A	Data-Integrity Audits	39
B	Supplementary Label-Collapse Comparisons	42
	References	43

List of Tables

2.1	Comparison framework for literature-review routes	5
3.1	Data sources, principal fields, and roles in the study	12
3.2	Data inclusion and exclusion rules. The table reports the processing stage, counting unit, operational condition, design rationale, remaining building count, and remaining image count	16
3.3	Vision configurations used for building-attribute prediction	18
3.4	Operational mapping from EP-Online building type (Gebouwtype) to TABULA-NL size class	22
4.1	City composition and fixed data partition of the experimental dataset	25
4.2	Upstream attribute extraction on the hold-out set ($n = 2,018$)	29
4.3	Joint-cell assignment on the hold-out set (majority-cell baseline = 0.790)	29
4.4	Binary downstream route comparison on the common hold-out set ($n = 2,018$)	31
4.5	Seven-class performance and ordinal error structure on the common hold- out set ($n = 2,016$)	32
4.6	Physical-consequence metric on the hold-out set (WIP; WWR = 0.25; rank- ing unchanged across 0.15/0.25/0.35)	32
4.7	Upstream degradation from pooled evaluation to LOCO (WIP; frozen DINOv2)	33
4.8	LOCO downstream route comparison (WIP; test set = Amsterdam, $n = 8,011$)	33
4.9	Attribute leave-one-out analysis and cumulative clean geometry (WIP; five- fold development OOF; baseline S_{full} macro-F1 = 0.181)	34
4.10	Extractability $\times$ utility (WIP; L1: frozen-embedding probe OOF R^2 ; L2: ground-truth injection $\Delta F1$)	34
4.11	Label-quality audit (WIP; manifest buildings and all raw EP-Online certificates)	35
4.12	Decomposition of pool-composition interventions (WIP; fixed feature set and hyperparameters; development baseline macro-F1 = 0.170, $\kappa = 0.184$, accuracy = 0.345)	35
A.1	Audit items and actions	40
A.2	Definitions of derived 3DBAG fields (corresponding to the source- characterisation note in section 3.2.1)	41
B.1	Literature-aligned label-collapse results	42

List of Figures

3.1	End-to-end study workflow.	10
3.2	Three-layer semi-synthetic dataset structure.	11
3.3	SVI acquisition and processing workflow.	14
3.4	Representative street-view crops by city and size class.	16
3.5	Vision-model configurations and multi-image aggregation.	20
3.6	TABULA-NL residential typology matrix.	21
4.1	Experimental-sample attribute and EPC-label distributions by city.	26
4.2	Per-cell recall heatmap for the 24 TABULA-NL archetype cells	30

Chapter 1

Introduction

1.1 Background and Motivation

The building sector accounts for approximately 40% of final energy consumption in the European Union and generates approximately 36% of its energy-related greenhouse gas emissions. The 2024 recast of the Energy Performance of Buildings Directive (EPBD) further established the policy objective of transforming the existing building stock into a zero-emission building stock by 2050 ([European Parliament and Council of the European Union, 2024](#)). However, the actual coverage of Energy Performance Certificate (EPC) data remains incomplete. The Netherlands introduced its EPC scheme in 2008, but by the end of 2019, only approximately 3.8 million residential buildings had a valid, registered EPC, representing around 48% of the residential building stock ([Concerted Action EPBD, 2020](#)). Large-scale model-based estimation can therefore complement formal energy certification for building-stock analysis and preliminary retrofit prioritisation, although its use should be clearly distinguished from legally issued EPC certification.

Urban Building Energy Modelling (UBEM) provides a bottom-up framework for analysing the operational energy use of tens to thousands of buildings at neighbourhood or city scale ([Reinhart & Cerezo Davila, 2016](#)). Because collecting detailed information on the geometry, construction, systems, and use of individual buildings is prohibitively costly, many UBEM workflows employ an archetype-based approach. Buildings with similar characteristics are grouped into categories, and each building inherits the thermal and system parameters associated with its archetype. TABULA is a representative European building-typology framework that defines residential typologies for 20 European countries according to building size, construction period, and other country-specific parameters ([Loga et al., 2016](#)). In the TABULA-NL configuration adopted in this study, building type and the construction period derived from construction year jointly determine the archetype cell, while floor count serves as an additional feature in the subsequent energy classification. Consequently, the large-scale application of archetype-based UBEM requires not only building geometry but also semantic attributes that are frequently missing, including building type, construction year, and floor count.

Previous research has shown that archetype lookup provides a consistent representation of the building stock but cannot capture all building-specific variation. Using approximately 2.57 million Dutch residential EPC records, [Hettinga et al. \(2023\)](#) evaluated large-scale energy labelling and reported an accuracy of 26% for TABULA rule-based assignment, below the majority-label baseline of 33%. By contrast, a random forest incorporating additional building-level and neighbourhood-level open-data features achieved 71% ac-

curacy. Because these results were obtained using different output-label ranges and evaluation sets, their absolute values should not be interpreted as a directly comparable model benchmark. Nevertheless, they indicate that assigning energy labels solely through typology rules has limited ability to capture variation at the individual-building level.

Recent advances in remote sensing and computer vision offer another means of obtaining missing building attributes. [Wurm et al. \(2021\)](#) used aerial imagery to infer construction type and construction period and incorporated these attributes into urban heat-demand modelling. [Mayer et al. \(2022, 2023\)](#) used street-view and aerial imagery to predict building energy-efficiency classes directly in the United Kingdom and Denmark, respectively. OpenFACADES, meanwhile, demonstrated the feasibility of large-scale, multi-attribute building-profile enrichment using street-view imagery (SVI) and vision-language models ([Liang et al., 2025](#)).

However, previous studies have primarily evaluated either image-based attribute extraction or direct image-to-energy classification in isolation, leaving attribute-mediated substitution without controlled evaluation. Specifically, it remains unclear which information is retained when building type, construction year, and floor count inferred from SVI replace reference attributes within the same TABULA-grounded workflow; which of these inferred attributes provide useful information for archetype assignment and downstream EPC classification; and what performance costs and applicability boundaries arise from this substitution. This gap motivates the present study and leads to the three research objectives presented in the following section.

1.2 Research Objectives

The large-scale application of archetype-based UBEM depends on the key metadata required to map individual buildings to appropriate archetypes. Existing open geospatial datasets, however, primarily provide building geometry, while semantic attributes such as building type, construction year, and floor count are frequently missing. Within the TABULA-NL framework, building type and the construction period derived from construction year determine the archetype cell, while floor count serves as an additional feature for subsequent energy classification.

The overall objective of this study is to evaluate whether building attributes inferred from street-view imagery (SVI) can replace the reference attributes provided by administrative and three-dimensional reconstruction data within a TABULA-based building-stock enrichment workflow, and to define the conditions under which such substitution is appropriate. To achieve this overall objective, the study establishes three specific research objectives:

1. **Evaluate SVI-based building-attribute extraction.** Assess the performance of different visual-inference configurations in predicting building type, construction year, and floor count from SVI.

2. **Determine the task-specific utility of the extracted attributes.** Determine the contribution of each SVI-derived attribute to TABULA archetype assignment and downstream EPC classification.
3. **Quantify the consequences and boundary conditions of attribute substitution.** Quantify the performance effects of replacing the full set of reference attributes with SVI-derived attributes and define the conditions under which such substitution is appropriate.

To support these objectives, this study constructs a three-layer, value-level traceable, TABULA-grounded semi-synthetic building dataset. The dataset integrates reference observations, SVI-derived attributes, deterministically assigned TABULA archetype parameters, and EP-Online EPC reference labels, and demonstrates its analytical use through a downstream EPC-classification case study.

Chapter 2

Literature Review

2.1 Comparison Framework

For the large-scale application of archetype-based UBE, previous studies can be organised into four workflow routes according to how building information is transferred from observational sources to energy-related tasks: the reference-attribute, proxy-attribute, direct visual, and attribute-mediated visual routes. These routes rely on different forms of data availability and differ in whether they retain explicit building attributes and the TABULA archetype structure. [table 2.1](#) establishes a common comparison framework based on the observed input, intermediate representation, typical target, and principal dependency of each route.

Table 2.1: Comparison framework for literature-review routes

Route	Observed input	Intermediate representation	Typical target	Principal dependency
Reference-attribute	Registry, geometry, and neighbourhood data	Recorded building attributes	EPC or energy-performance prediction	Complete and linkable building-level records
Proxy-attribute	Aggregated census, morphology, or remote-sensing data	Inferred attributes or archetype classes	Archetype assignment or energy modelling	Transferable relationship between the proxy and the individual building
Direct visual	SVI, aerial, or multimodal imagery	Latent visual representation	Direct EPC classification	Local labels, image coverage, and domain stability
Attribute-mediated visual	SVI	Explicit building type, construction year, floor count, and TABULA cell	Archetype parameters and downstream EPC classification	Attribute extractability and error propagation

The studies considered here differ in their label spaces, source populations, data splits, and metrics. Their numerical results are therefore used to characterise the capabilities and boundaries of each route, rather than to rank performance directly across studies.

2.2 Reference and Open-Data Approaches

When complete building-level records are available, the reference-attribute route uses administrative, geometric, and neighbourhood data to train statistical models. [Hettinga et al. \(2023\)](#) developed an EPC classifier for Dutch residential buildings by combining BAG building year, building type, ground-floor area, total floor area, and exposed-envelope geometry with CBS neighbourhood-level home ownership and average building value. Their random forest achieved 71% accuracy and 93% ± 1 -grade accuracy on the held-out test subset of an 80/10/10 train–validation–test protocol. By comparison, TABULA rule-based assignment achieved 26% accuracy and 67% ± 1 -grade accuracy across 2,574,227 EPC records. The two approaches used different evaluation sets, and the EPC reference labels covered A++–G whereas the TABULA outputs covered only B–G. These values therefore demonstrate the analytical potential of additional open-data features, rather than providing a fully equivalent model comparison.

This route retains interpretable tabular inputs and can incorporate building-level and neighbourhood-level information beyond the archetype. However, it assumes that the target region already has linkable reference attributes of sufficient quality. When building type, construction year, or floor count is missing, the route alone cannot determine how these inputs should be reconstructed or quantify the cost of replacing them with information from other sources.

2.3 Proxy-Based Attribute and Archetype Inference

The proxy-attribute route uses data sources that correlate with a target attribute but do not constitute observations of the individual building itself. [Dabrock et al. \(2025\)](#) combined German census data aggregated to a 100 m $\times$ 100 m grid with building-level and neighbourhood-level morphological and socioeconomic features, and trained XG-Boost models to predict residential size class and construction-year class. The two tasks achieved accuracies of 97.4% and 73.9%, respectively, on an individual-building test set, and the predictions were subsequently used to construct a nationwide TABULA-based building dataset. The study also observed that the resulting dataset tended to overrepresent majority classes, indicating that aggregate proxies can support individual-building estimates but that their errors are affected by grid composition and class imbalance.

Remote sensing can also serve as an attribute proxy. [Wurm et al. \(2021\)](#) extracted construction type and construction period from aerial imagery and linked these attributes to urban heat-demand modelling. Their study demonstrated that imagery-derived attributes can be incorporated into a downstream energy model. However, the imagery source was

not SVI, and reference attributes and image-derived substitutes were not compared for the same buildings under a common downstream protocol. Proxy-based methods therefore offer a feasible route for large-scale data completion, but their transferability remains dependent on proxy resolution, geographic context, and target taxonomy.

2.4 Direct Image-to-EPC Approaches

The direct visual route bypasses explicit archetype assignment and predicts EPC labels directly from imagery. [Mayer et al. \(2023\)](#) integrated SVI, aerial imagery, building footprints, and land-surface temperature for approximately 40,000 buildings in the United Kingdom and performed binary classification of A–D versus E–G. The complete multimodal model achieved a macro-F1 score of 64.64%. In a study of more than 21,000 buildings in the Copenhagen metropolitan area, the same research group obtained a macro-F1 score of 58.79% using SVI and aerial imagery, and concluded that these two image sources alone were insufficient to distinguish building energy efficiency consistently ([Mayer, Heilborn, & Fischer, 2023](#)).

Subsequent studies expanded both the data sources and label granularity. [Liu et al. \(2025\)](#) combined Google Street View and GIS data in a three-stage framework using 1% local samples for transfer learning. They reported an accuracy of 84% for A–G EPC classification, although performance decreased for occluded high-rise buildings. [Sun et al. \(2026\)](#) integrated thermal infrared imagery, optical remote sensing, SVI, morphology, and socioeconomic indicators. Their binary high- versus low-performance tasks achieved F1 scores of 0.64 in Glasgow and 0.69 in Edinburgh. In their ablation study, the SVI-only model achieved F1 scores of 0.59 and 0.68 in the two cities, respectively.

These findings support the presence of EPC-predictive signals in building appearance and urban-context data, but the capability of the direct visual route cannot be summarised by a single performance value. Previous studies differ substantially in modality, local adaptation, label aggregation, class definition, and evaluation protocol. Most employ binary targets, while only a small number report A–G classification; consequently, a directly comparable full-grade benchmark has not yet emerged. Moreover, direct models bypass explicit TABULA parameters. Although they can directly support classification, they cannot independently identify the contribution of individual building attributes or directly quantify how attribute substitution affects the archetype cell and its inherited physical parameters.

2.5 Image-Based Building-Attribute Extraction

The attribute-mediated visual route first infers explicit building attributes from SVI and then performs TABULA cell assignment or downstream modelling. In contrast to the direct route, this structure retains intermediate attributes and thereby enables separate evaluation of attribute extraction, joint-cell assignment, and downstream utility. The relevant vision

strategies can be further divided into supervised full fine-tuning, frozen self-supervised representations, and inference without parameter updating.

2.5.1 Supervised Full Fine-Tuning

OpenFACADES compared CNN, Vision Transformer, and VLM configurations using 31,180 labelled images from seven cities ([Liang et al., 2025](#)). Fine-tuned InternVL3-2B achieved an accuracy of 61.27% for building-type classification. For building-age regression, it achieved an R^2 of 0.710 and an MAE of 28.06 years, while the corresponding values for floor-count regression were 0.789 and 2.13 floors. The study also applied the model to large-scale building-profile enrichment, demonstrating that a single image-processing framework can support multi-attribute inference. Full fine-tuning, however, still requires a labelled training corpus, and its applicability across cities or taxonomies must be established through independent transfer evaluation.

2.5.2 Frozen Self-Supervised Representations

Self-supervised visual foundation models learn general-purpose representations from large-scale image corpora without manual annotation. DINOv2 provides visual features that can be used with task-specific prediction heads without updating the backbone parameters ([Oquab et al., 2023](#)). Compared with full fine-tuning, a frozen-backbone configuration reduces the number of parameters that must be updated and separates representation learning from downstream supervised fitting. However, the building-energy studies reviewed above do not provide controlled evidence that compares a frozen self-supervised backbone with a fully fine-tuned CNN and VLM inference on the same building population and traces their attribute errors through TABULA cell assignment to EPC classification.

2.5.3 VLM Inference Without Parameter Updating

VLMs offer another route for building-attribute inference without updating model parameters. [Zeng et al. \(2024\)](#) used GPT-4 Vision for zero-shot classification of building-age epochs from London façade images, achieving an accuracy of 39.69% and an epoch-class MAE of 0.85. The model was less reliable for very old buildings and for fine-grained distinctions between periods separated by no more than two decades. OpenFACADES also compared zero-shot and fine-tuned VLM configurations, with its overall results showing that fine-tuning improved multi-attribute performance ([Liang et al., 2025](#)). Inference without parameter updating therefore reduces the need for task-specific training, but its results remain sensitive to model access, prompt design, output constraints, and geographic context. Furthermore, the term “zero-shot” does not necessarily mean that development examples played no role in prompt selection.

2.6 Archetype-Grounded and Semi-Synthetic Datasets

The routes reviewed above also produce different forms of building-level datasets. [Dabrock et al. \(2025\)](#) mapped predicted size and construction-year classes to TABULA archetypes to construct a nationwide residential dataset. OpenFACADES, by contrast, extended image-derived attributes and captions to approximately 1.2 million images covering 500,000 buildings ([Liang et al., 2025](#)). These studies demonstrate archetype instantiation and image-based data enrichment, respectively, but neither organises reference attributes, image-derived substitutes, deterministically assigned TABULA parameters, and EPC reference labels within a single paired-comparison structure.

The recent Building Energy Graphs (BEGs) preprint further combined OpenStreetMap footprints, GHSL products, European building archetypes, and deterministic assumptions to construct simulation-ready building energy models, with EnergyPlus outputs supporting GNN surrogate prediction ([Dalach et al., 2025](#)). This work demonstrates that observed geospatial sources, archetype assumptions, and simulation outputs can be integrated into synthetic or semi-synthetic learning artefacts. Its target, however, is simulated energy-use intensity rather than administrative EPC reference labels. This distinction positions the utility of the dataset developed in the present study as measurement-anchored attribute-substitution and EPC analysis, rather than as another simulation-surrogate benchmark.

Chapter 3

Method

3.1 Overview

3.1.1 Pipeline Architecture

The pipeline developed in this study is designed to examine how replacing measured or administrative building attributes with attributes inferred from street-view imagery, and subsequently converting those inferred attributes into U-values through a fixed TABULA/archetype lookup, affects downstream EPC classification and physical consistency. Data processing comprises four sequential stages. First, street-view images are paired with the target buildings, while the image source, view quality, and pairing status are retained. Second, eligible images are used to infer building type, construction year, and floor-related attributes, with the predicted values and unreadable cases recorded separately. Third, only the attributes required by the lookup key are used to deterministically assign each building to a TABULA/archetype cell, from which the substituted thermal parameters and their provenance are obtained. Fourth, the features required by the downstream model are assembled for each experimental route, and the physical parameters required to calculate H'_{tr} are prepared.

The evaluation design is defined separately from the preceding data-processing stages. M1 uses reference attributes or reference thermal parameters, whereas M3 replaces the corresponding inputs with attributes inferred from street-view imagery and the results of the associated archetype lookup; together, these routes form the primary substitution comparison. M2 instead predicts EPC directly from imagery and serves as a parallel end-to-end comparator rather than an intermediate stage of the archetype pipeline. All comparable routes use the same geographic split. Binary EPC classification is the primary task, while seven-class EPC prediction is used to diagnose the information ceiling associated with fine-grained labels. LOCO evaluates cross-location generalisation, and H'_{tr} provides a physical consistency check for the substituted thermal parameters.

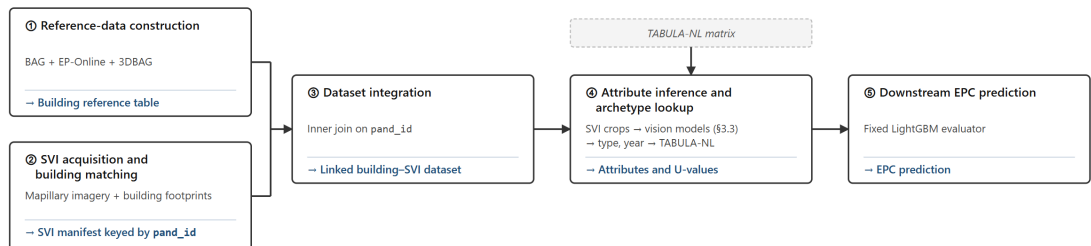


Figure 3.1: End-to-end study workflow.

3.1.2 Semi-Synthetic Dataset: Three-Layer Structure

This study defines the semi-synthetic dataset as three layers with explicitly separated provenance. First, the reference layer contains administrative records from BAG and EP-Online, reconstructed geometry from 3DBAG, and the EPC reference label adopted in this study. These fields provide the reference observations required for model training, attribute substitution, and downstream evaluation, but they are not collectively treated as error-free ground truth. In particular, the EPC reference label remains affected by multiple certificates and within-building label disagreement; its quality is audited separately in [section 4.3.4](#).

Second, the vision-inference layer stores the building-level predictions of building type, construction year, and floor count produced by the three vision paradigms, together with their model provenance. Third, the synthetic archetype layer derives the construction period from the estimated year, deterministically maps the pair (type, period) to a TABULA-NL cell, and assigns the cell's four as-built envelope U-values for the wall, roof, floor, and windows. These U-values are archetype-level average assumptions for the corresponding building-stock subgroup; they are not measurements of the thermal performance of individual buildings and do not introduce new building-specific observations.

Here, semi-synthetic refers to the combination of reference observations, image-derived attributes, and deterministically assigned archetype parameters. The EPC target remains an administrative reference label obtained from EP-Online rather than a simulated energy label; only the archetype parameters assigned through the TABULA cell are synthetic. This separation makes the provenance of each value traceable and provides the common data foundation for the three research objectives.

In the controlled substitution comparison, M1 and M3 use the same eligible buildings, geographic split, feature positions, and fixed model for each downstream task. M1 uses the reference type, year, and floor attributes and obtains the TABULA U-values from the corresponding reference (type, period) cell. M3 replaces these inputs with the type, year, and floor attributes inferred from SVI and retrieves the U-values from the resulting predicted cell. Within this protocol, the difference between M3 and M1 therefore measures the consequence of substituting the attribute source rather than the effect of retraining the downstream model. M2 predicts EPC directly from imagery and does not pass through this three-layer substitution chain; it is consequently treated as a parallel end-to-end comparator.

KEY pand_id	REFERENCE LAYER		VISION-INFERENCE LAYER		SYNTHETIC ARCHETYPE LAYER	
	Reference attributes	EPC label	Vision model	Vision-inferred attributes	M1: reference-derived TABULA	M3: vision-derived TABULA
0363100012162533	AB, 1883, 3 floors	C	DINOv2	AB, 1903, 4.5 floors	AB · NL.01 (1.6129, 1.5385, 1.7241, 5.2)	AB · NL.01 (1.6129, 1.5385, 1.7241, 5.2)
0599100000752878	TH, 2009, 3 floors	A	DINOv2	TH, 2004, 3.6 floors	TH · NL.05 (0.266, 0.2347, 0.266, 1.8)	TH · NL.04 (0.3584, 0.3584, 0.3584, 2.9)
0363100012074053	AB, 1989, 3 floors	B	DINOv2	TH, 1981, 2.9 floors	AB · NL.03 (0.641, 0.641, 1.2821, 5.2)	TH · NL.03 (0.641, 0.641, 1.2821, 5.2)
0344100000077356	AB, 2009, 3 floors	A	DINOv2	TH, 1924, 2.8 floors	AB · NL.05 (0.266, 0.2347, 0.266, 1.8)	TH · NL.01 (2.2222, 2.0833, 2.439, 5.2)

Figure 3.2: Three-layer semi-synthetic dataset structure.

3.2 Dataset Construction

3.2.1 Sources

The dataset is constructed around the BAG `pand_id` as the building-level identifier and integrates five public or openly licensed data sources: BAG, 3DBAG, EP-Online, Mapillary, and TABULA-NL. The building-level linkage of BAG, EP-Online, and Mapillary forms the core sample; 3DBAG provides the reconstructed geometry used in subsequent stages, while TABULA-NL provides a deterministic archetype lookup based on building type and construction period. The position of each source within the three-layer data structure is shown in [fig. 3.2](#), and the principal fields and their roles in the study are summarised in [table 3.1](#).

Table 3.1: Data sources, principal fields, and roles in the study

Source	Main contents	Role in this study
BAG	<code>pand_id</code> , bouwjaar, two-dimensional building footprints, and building status	Building-level join key, construction-year reference, and residential-stock filtering
3DBAG	LoD2.2 reconstructions of floor count, height, volume, and envelope areas	Floor-count reference, Tier B reconstructed geometry, and area inputs for H'_{tr}
EP-Online	Energieklasse, Gebouwtype, and Registratiedatum	EPC reference label, building-type reference, and certificate selection by date
Mapillary	Crowdsourced street-view imagery and image metadata	Vision-model input, SVI–building pairing, and image provenance
TABULA-NL	Six construction periods $\times$ four size classes, comprising 24 archetype cells	Deterministic lookup from (type, period), providing as-built U-values for walls, roofs, floors, and windows

Mapillary was selected instead of a commercial street-view service because its open licensing supports a reproducible research workflow. However, its image coverage is determined by contributor activity and does not constitute a random sample of the building stock. The resulting dataset may therefore be biased towards image-dense urban centres and particular residential types. Representative cropped street-view images are shown in [fig. 3.4](#); the coverage bias is quantified through the sample distributions in [section 4.1.1](#), and its implications for the scope of the study are discussed in [section 5.2](#).

The downstream labels are constructed directly from the EP-Online `Energieklasse`, with ratings from A+ to A++++ first merged into class A. The primary task is an independently trained binary EPC classification task that groups the labels into A–C and D–G; the

division between the two groups corresponds to the upper boundary of EPC class C at 250 kWh/(m²·yr). Seven-class A–G prediction is retained as a diagnostic task for examining the information ceiling associated with fine-grained labels. Both tasks are derived directly from the registered *Energieklasse*; continuous-energy regression is an auxiliary analysis and is not used to define the primary labels.

3.2.2 SVI Acquisition, Data Linkage, and Screening Rules

SVI acquisition and image–building pairing. This study uses a locally adapted version of the OpenFACADES framework (Liang et al., 2025) to acquire and process Mapillary street-view imagery; the complete workflow is shown in fig. 3.3. OpenFACADES first acquires and integrates building footprints from OpenStreetMap and Overture Maps. In this study, the workflow is divided into separate footprint-acquisition and image-processing stages, with BAG-based residential pre-filtering introduced between them so that footprints outside the target residential stock do not enter the subsequent image-processing workflow.

Residential pre-filtering is performed using the geometric centroid of each footprint. The OpenFACADES footprints and BAG building polygons are first transformed to the Dutch national coordinate reference system EPSG:28992, after which the geometric centroid of each OpenFACADES footprint is calculated. A footprint is retained only when its centroid falls within a BAG residential-building polygon. This preliminary step determines only which footprints proceed to image processing.

For the retained footprints, OpenFACADES acquires metadata for Mapillary 360° panoramic images within the study area and performs isovist analysis using a 30 m search radius to establish candidate pairings between panoramic images and visible buildings. Following angle-of-view (AoV) screening, the candidate images are downloaded at a resolution of 2,048 pixels. GroundingDINO is then used to detect the target building, and perspective reprojection and cropping are applied to produce single-building images before the final image-quality filtering stage.

Each cropped image retains its OpenFACADES `building_id`, whereas the building attributes and energy labels in this study use the BAG `pand_id` as their primary key. After image processing, the same centroid-based matching rule is therefore applied to create the formal crosswalk between the OpenFACADES `building_id` and BAG `pand_id`. When the centroid of a footprint falls within a BAG residential-building polygon, the building's `pand_id` is assigned to the corresponding images. Images for which no BAG match can be established are excluded.

Building-level reference table and data linkage. The building reference data and SVI manifest are constructed separately. The EP-Online data are first restricted to residential categories and existing-building certificates; where multiple records share the same BAG building identifier, only the record with the most recent registration date is retained. BAG, 3DBAG, and EP-Online are then joined using an inner join on `pand_id`, after which TABULA-NL archetype lookup is performed based on building type and construction period

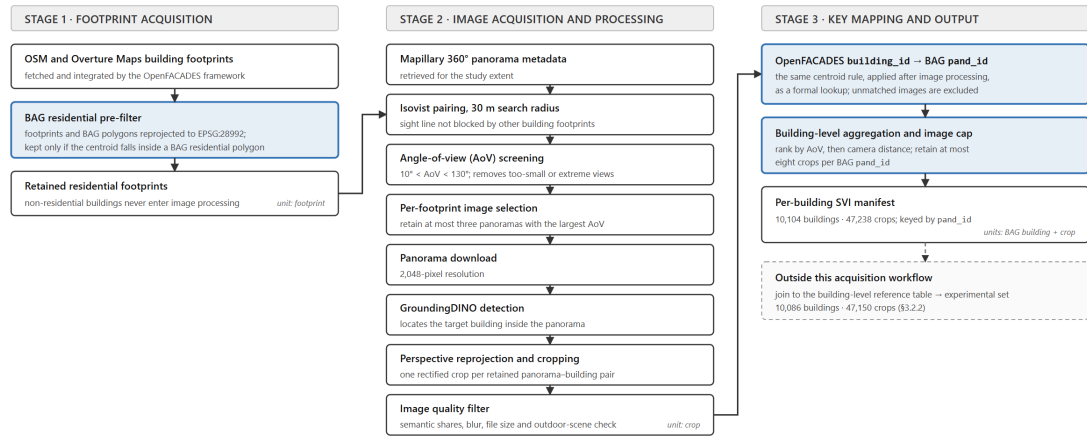
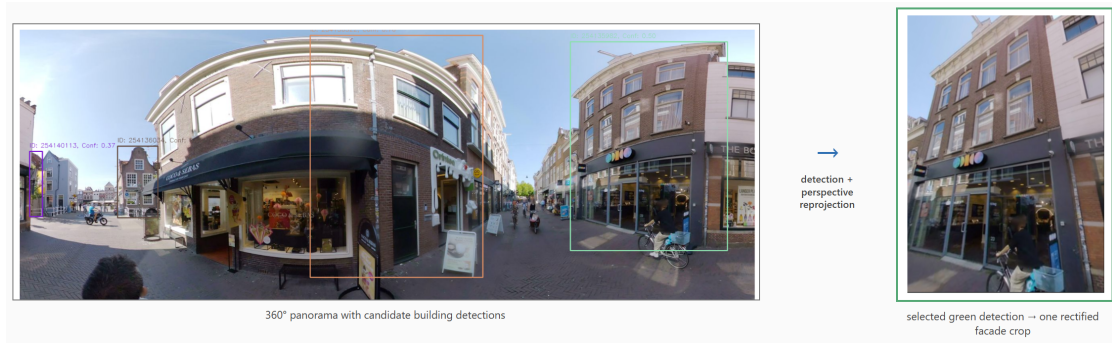


Figure 3.3: SVI acquisition and processing workflow.



Panorama detection and facade-crop reprojection.

to form the building-level reference table. Finally, the SVI manifest and the TABULA-matched reference table are joined using an inner join on `pand_id`. The label uncertainty introduced by building-level aggregation based on the most recent certificate is examined further in [section 4.3.4](#).

Records are selected and quality-controlled according to the following rules:

- 1. Residential and existing buildings:** The BAG usage designation must include `woonfunctie`. EP-Online records must belong to the residential category (`Gebouwklasse = W`), have existing-building status (`Status = Bestaand`), and contain a non-missing `Energieklasse`. These conditions restrict the study scope to residential buildings and ensure compatibility with the TABULA-NL residential typology.
- 2. Reference attributes and archetype matching:** Only buildings with `1800 <= bouwjaar <= 2025` are retained, thereby excluding missing, anomalous, or out-of-period construction years. The EP-Online `Gebouwtype` must map to one of the four residential types used in this study, the estimated floor count must lie between 1 and 30, and the building must permit TABULA-NL archetype matching.
- 3. EPC record selection and methodological period:** When the same BAGPandIDs value is associated with multiple EP-Online certificates, the record with the most recent `Registratiedatum` is retained. The study does not explicitly filter for NTA

8800 certificates using `Berekeningstype`. However, in the EP-Online snapshot dated 1 April 2026, all certificates retained in the final sample were registered after 2021 and therefore fall within the period in which NTA 8800 was in effect.

4. **Geometric image–building pairing:** The Mapillary camera position must fall within a 30 m search radius of the target building, and an unobstructed geometric line of sight must exist between the camera and the building without intersection by another building footprint. Candidate pairings must also satisfy $10\text{ deg} < \text{AoV} < 130\text{ deg}$ to exclude images in which the building is too small or the viewing angle is excessively oblique. This geometric assessment does not account for trees, vehicles, or other temporary obstructions.
5. **Image-quality filtering (heuristic):** A cropped image must satisfy all of the following conditions: a pixel proportion of at least 15% for the `Building` class, a pixel proportion of no more than 30% for the `Wall` class, Laplacian variance greater than 30, a file size of at least 20,000 bytes (approximately 20 kB), and an outdoor scene classification from `Places365`. These thresholds exclude images with targets that are too small, unsuitable visual content, severe blur, or anomalous files.
6. **Maximum image count:** Image counts are limited at two levels. During OpenFACADES image selection, a maximum of three panoramic images with the largest AoV values is retained for each OpenFACADES footprint. Because multiple OpenFACADES footprints may correspond to the same BAG `pand_id`, a BAG building may still have more than three cropped images after identifier linkage. Images associated with the same `pand_id` are therefore sorted by decreasing AoV and, where AoV values are equal, by increasing camera-to-building distance. A maximum of eight images is ultimately retained for each BAG building.

The inclusion and exclusion rules and their corresponding sample counts are summarised in [table 3.2](#). Because the workflow contains several distinct counting units, including OpenFACADES footprints, panorama–footprint pairs, cropped images, and BAG buildings, [table 3.2](#) reports the building and image counts separately at each stage rather than combining different units into a single funnel count.

After image-quality filtering, `pand_id` linkage, and application of the eight-image-per-building limit, the SVI manifest contains **10,104 buildings and 47,238 images**. When the manifest is linked to the building-level reference table, 88 images associated with 18 buildings are excluded from the experimental dataset because of invalid 3DBAG geometry or unsuccessful TABULA-NL archetype matching. The final experimental dataset therefore contains **10,086 buildings and 47,150 images**.

The Mapillary image-acquisition dates and EPC registration dates are not temporally matched or filtered in this study. The potential temporal mismatch between images and labels is discussed in [section 5.2](#). Representative cropped street-view images are shown in [fig. 3.4](#).

Table placeholder
Data inclusion and exclusion rules. The table reports the processing stage, counting unit, operational condition, design rationale, remaining building count, and remaining image count

Table 3.2: Data inclusion and exclusion rules. The table reports the processing stage, counting unit, operational condition, design rationale, remaining building count, and remaining image count

City	SFH Single-Family House	TH Terraced House	MFH Multi-Family House	AB Apartment Block
Amsterdam 8,019 bld / 36,569 img				
Rotterdam 1,424 bld / 7,366 img				
Utrecht 488 bld / 2,233 img				
Delft 173 bld / 1,070 img			no MFH in the Delft subset	

Figure 3.4: Representative street-view crops by city and size class.

3.2.3 Data Partitioning and Cross-Validation

The final experimental dataset is partitioned in two stages at the building level. In the first stage, five-fold `StratifiedGroupKFold` is applied to the 10,086 buildings, with one fold retained as a 20% hold-out set comprising 2,018 buildings and the remaining 80% used as a development set comprising 8,068 buildings. The stratification variable combines the study area (`city`), the original EP-Online residential-type field (`Gebouwtype`), the registered EP-Online energy-class field (`Energieklasse`), and the derived TABULA construction period (`tabula_period`). `BAG pand_id` is used as the grouping key to ensure that multiple images of the same building cannot be distributed across different data subsets.

In the second stage, five-fold cross-validation is performed within the development set using `StratifiedGroupKFold` again. For every trainable model, model fitting, hyperparameter selection, and model selection use only the development set. The frozen hold-out set does not participate in model selection and is used only for final evaluation. All comparison routes use the same data partition to ensure direct comparability of the model results.

3.3 Vision Inference Strategies and Training Configurations

Three open-source vision-model configurations are specified in advance to represent full-model fine-tuning, supervised prediction-head training on a frozen self-supervised backbone, and vision-language model inference without parameter updating. The configurations differ simultaneously in model architecture, pre-training method, parameter-update scope, number of images used per building, and multi-image aggregation. The study therefore compares three complete and reproducible inference configurations rather than using a factorial experiment to isolate the causal effect of any individual training strategy. Closed-source commercial models are excluded because their model versions, access conditions, and long-term reproducibility are more difficult to control.

Table 3.3: Vision configurations used for building-attribute prediction

Configuration	Pre-trained weights	Parameters updated on the development set	Development labels used	Images per building	Multi-image aggregation
ResNet-50 full fine-tuning	ImageNet-pre-trained ResNet-50	Backbone, shared MLP, and three attribute-output heads	Complete development labels	Up to 8	Image features averaged before building-level prediction
Frozen DINOv2 ViT-B/14	vit_base_patch14_-dinov2.lvd142m	Shared MLP and three attribute-output heads only; backbone frozen	Complete development labels	Up to 8	Image features averaged before building-level prediction
InternVL3-2B without parameter updating	OpenGVLab/InternVL3-2B	None	Fixed, class-balanced subset of 200 development buildings used for prompt design	Up to 3	Mode for building type; median for construction year and floor count

Supervised attribute prediction. ResNet-50 and DINOv2 both use up to eight images per building, each resized to 224×224 pixels. Each image is first transformed into a feature vector by the same backbone, and the resulting vectors are averaged arithmetically at the building level. The mean feature vector passes through a shared 256-dimensional MLP, after which three separate output heads predict Gebouwtype, construction year, and floor count. The attribute-training objective is

$$\mathcal{L}_{\text{attr}} = \mathcal{L}_{\text{CE,type}} + \mathcal{L}_{\text{MSE,year}} + \mathcal{L}_{\text{MSE,floor}}.$$

Building type uses cross-entropy loss weighted inversely by class frequency in the corresponding development training fold. Construction year and floor count are first normalised using the mean and standard deviation of that training fold, after which mean squared error is calculated. The three loss terms are summed with equal weights, which are not adjusted using hold-out data. ResNet-50 updates all parameters in the backbone, shared MLP, and three output heads; DINOv2 freezes the backbone and updates only the shared MLP and output heads. Both configurations are trained in five folds within the development set. Early stopping in each fold is based on the validation Gebouwtype macro-F1, and final hold-out evaluation uses the fold checkpoint with the highest development-validation Gebouwtype macro-F1.

VLM-based attribute inference. InternVL3-2B does not update model parameters using the study data. Instead, deterministic decoding returns JSON containing `building_type`, `construction_year`, and `num_floors`. The prompt wording and output rules are designed using a fixed, class-balanced subset of 200 development buildings and then frozen before hold-out evaluation. Accordingly, “zero-shot” in this study means only that the model parameters are not trained or fine-tuned using the study data; it does not mean that the overall analytical workflow makes no use of development labels. Up to the first three ranked images are used for each building. Parseable image-level responses are aggregated at the building level: building type is determined by the mode, construction year by the median, and floor count by the rounded median. Responses that cannot be parsed or fall outside the permitted ranges are marked as invalid and excluded from aggregation.

Direct EPC prediction (M2). In addition to the building-attribute models used for the M3 route, direct energy-label prediction configurations are established for the M2 route. For ResNet-50 and DINOv2, the three attribute-output heads are replaced by a single energy-classification head. The seven-class *Energieklasse* task and the binary A–C/D–G task are trained independently using cross-entropy loss weighted inversely by class frequency in the corresponding development training fold. ResNet-50 undergoes full-model fine-tuning for M2. DINOv2 instead uses pre-extracted and cached building-level mean features and trains only the energy-classification network. Checkpoint selection within each fold and selection of the final fold are both based on development-validation energy-label macro-F1. InternVL3-2B uses separate direct-prediction prompts for the seven-class and binary tasks. Each image produces one class response, and the final prediction is obtained by majority voting at the building level.

Because the two supervised configurations use up to eight images with feature-level aggregation, whereas InternVL3-2B uses up to three images with prediction-level aggregation, their results should be interpreted as comparisons between complete model configurations. The independent effects of image count and aggregation method are outside the identification scope of this study.

Figure 3.5 summarises the input, parameter-update scope, multi-image aggregation, and prediction outputs of the three vision-inference configurations.

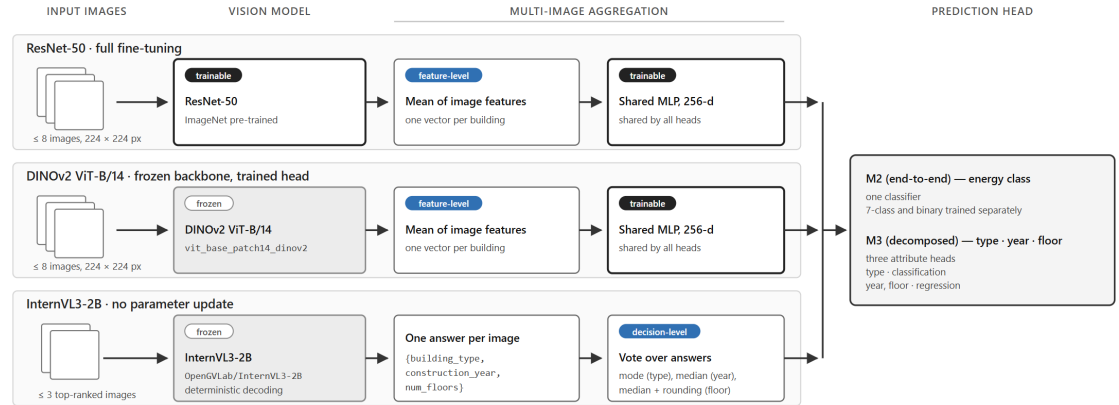


Figure 3.5: Vision-model configurations and multi-image aggregation.

3.4 TABULA Archetype Matching

3.4.1 NL Matrix Structure

The Dutch TABULA matrix consists of six construction periods $\times$ four size classes, forming 24 archetype cells in total (fig. 3.6). Each cell corresponds to a TABULA-NL archetype identifier that is subsequently used to retrieve the associated as-built thermal parameters. The period boundaries and operational definitions of the size classes are taken from the Dutch TABULA national typology report, the RVO *Example Dwellings* (*Voorbeeldwoningen*) series, rather than from Loga et al. (2016). The latter provides a methodological and cross-national comparison framework but does not contain the Dutch classification thresholds.

Size-class definitions. The four size classes are defined according to the Dutch dwelling types represented in the TABULA-NL matrix; no additional thresholds based on the number of dwelling units or floors are imposed:

- **SFH — Single-Family House:** Detached house (*Vrijstaande woning*) and semi-detached house (*Twee-onder-één-kap*).
- **TH — Terraced House:** Mid-terrace house (*Rijwoning tussen*) and end-terrace house (*Rijwoning hoek*).
- **MFH — Multi-Family House:** Residential building with non-self-contained accommodation (*Woongebouw met niet-zelfstandige woonruimte*).

	Single-family housing		Collective housing	
Construction period	Single-Family House (SFH) Vrijstaande woning / Twee-onder- één-kap	Terraced House (TH) Rijwoning tussen / hoek	Multi-Family House (MFH) Woongebouw (niet-zelfstandige woonruimte)	Apartment Block (AB) Appartement
≤ 1964 NL.01	 NL.N.SFH.01.Gen	 NL.N.TH.01.Gen	 NL.N.MFH.01.Gen	 NL.N.AB.01.Gen
1965–1974 NL.02	 NL.N.SFH.02.Gen	 NL.N.TH.02.Gen	 NL.N.MFH.02.Gen	 NL.N.AB.02.Gen
1975–1991 NL.03	 NL.N.SFH.03.Gen	 NL.N.TH.03.Gen	 NL.N.MFH.03.Gen	 NL.N.AB.03.Gen
1992–2005 NL.04	 NL.N.SFH.04.Gen	 NL.N.TH.04.Gen	 NL.N.MFH.04.Gen	 NL.N.AB.04.Gen
2006–2014 NL.05	 NL.N.SFH.05.Gen	 NL.N.TH.05.Gen	 NL.N.MFH.05.Gen	 NL.N.AB.05.Gen
≥ 2015 NL.06	 NL.N.SFH.06.Gen	 NL.N.TH.06.Gen	 NL.N.MFH.06.Gen	 NL.N.AB.06.Gen

Figure 3.6: TABULA-NL residential typology matrix.

- **AB — Apartment Block:** Apartment (Appartement).

3.4.2 Attribute-to-Archetype Mapping

The TABULA-NL archetype cell of each building is jointly determined by its building size class and construction period. Let s_i denote the TABULA size class of building i , and let $p(y_i)$ denote the construction period obtained by discretising construction year y_i according to the boundaries shown in [fig. 3.6](#). The archetype cell is then defined as

$$c_i = (s_i, p(y_i)).$$

In the reference-attribute route, construction year is obtained from the BAG bouwjaar, while the EP-Online building type (Gebouwtype) is converted deterministically to SFH, TH, MFH, or AB according to [table 3.4](#). In the vision-attribute route, the model predicts these four TABULA size classes directly and does not use the EP-Online Gebouwtype enumeration as an intermediate mapping. The predicted continuous construction year is discretised using the same period boundaries. Both routes use the same archetype lookup at this stage; they differ only in whether the input attributes originate from reference data or vision predictions.

Table 3.4: Operational mapping from EP-Online building type (Gebouwtype) to TABULA-NL size class

EP-Online building type (Gebouwtype)	TABULA-NL size class
Detached house (Vrijstaande woning)	SFH
Semi-detached house (Twee-onder-één-kap)	SFH
Mid-terrace house (Rijwoning tussen)	TH
End-terrace house (Rijwoning hoek)	TH
Residential building with non-self-contained accommodation (Woongebouw met niet-zelfstandige woonruimte)	MFH
Apartment (Appartement)	AB

[Table 3.4](#) defines the deterministic operational mapping adopted in this study. The reference-attribute route maps the EP-Online enumeration values directly and does not reclassify building type from street-view imagery. Records outside the six retained values listed above are excluded during construction of the building-level reference dataset. All archetype assignments use BAG pand_id as the unit of analysis and the retained EP-Online record defined in [section 3.2.2](#); the resulting unit-of-observation limitation is discussed in [section 5.1.2](#).

Floor count does not participate in archetype-cell assignment because the TABULA-NL lookup key consists only of size class and construction period. `num_floors` is nevertheless retained as a separate feature in the downstream EPC-classification model.

3.4.3 U-Value Lookup and As-Built Assumption

After archetype-cell assignment, the wall, roof, floor, and window U-values are retrieved from the TABULA-NL lookup table according to the building's size class and construction period. All four values are expressed in $W/(m^2K)$. The reference-attribute route performs the lookup using the reference building type and construction year, while the vision-attribute route applies the same procedure using the image-inferred building type and construction year. The U-values, element variants, and source records for each cell are provided in Appendix A.

Because renovation status is unavailable at the individual-building level, every building is assigned the unrenovated existing-state, or as-built, variant of its TABULA-NL cell. This is a modelling assumption rather than an observation of the building's current envelope condition. For thermally renovated buildings, the assigned U-values may exceed the current actual thermal transmittance and may therefore overestimate transmission heat loss.

TABULA-NL U-values represent archetype-level assumptions for the corresponding building-stock subgroup rather than measured thermal parameters for individual buildings (Loga et al., 2016). Buildings with the same size class and construction period therefore receive the same four U-values; these values cannot establish the actual envelope construction or renovation status of an individual building.

3.5 Downstream EPC Classification Model

A LightGBM gradient-boosting classifier (Ke et al., 2017) is used as the downstream EPC-classification model. LightGBM is suitable for low-dimensional tabular data comprising numerical and categorical variables and allows the same model architecture and training configuration to be maintained across different attribute sources. In this study, it serves as a controlled downstream evaluator for quantifying the effect of replacing reference attributes with image-inferred attributes on EPC prediction; it is not presented as a contribution of a new classification algorithm.

The model has eight core input fields: building type, construction year, floor count, wall U-value, roof U-value, floor U-value, window U-value, and city. Building type and city are treated as categorical features, while the remaining fields are numerical features. The reference-attribute route uses the reference building type, construction year, and floor count and obtains the four U-values from the corresponding TABULA-NL cell. The vision-attribute route instead uses the three image-inferred attributes and retrieves the U-values from the predicted TABULA-NL cell. The two routes therefore have the same

eight-field feature schema, and city remains unchanged; the only differences arise from the source of the three upstream attributes and their derived archetype U-values. Apart from the EPC target label and the fields used to construct the reference attributes, no other certificate-derived variables are used because they would be unavailable for target buildings without an existing energy certificate.

The seven-class A–G and binary A–C/D–G EPC tasks use two independently trained LightGBM classifiers with multiclass and binary objectives, respectively. Both tasks use the same fixed hyperparameter configuration, without route-specific tuning. The principal settings are 400 boosting iterations, a learning rate of 0.05, 31 leaves, and no class weighting; the complete configuration is provided in Appendix A.

The final downstream classifier is fitted once on the complete development set using reference features. The same fitted model is then applied separately to the reference-derived and vision-derived hold-out features. This protocol avoids introducing additional differences through downstream-model retraining, so that the prediction gap reflects the combined consequences of attribute substitution and propagation through the TAB-ULA lookup. The development evaluation, hold-out protocol, and evaluation metrics are described in [section 4.1](#).

Chapter 4

Experiments and Evaluation

This chapter presents the quantitative evaluation. [section 4.1](#) defines the experimental strategy, including dataset statistics, comparison-route families, and all evaluation metrics. [section 4.2](#) reports the quantitative results, while [section 4.3](#) presents the qualitative and diagnostic analyses. The finding statement accompanying each table describes only the observed pattern; causal interpretation is reserved for [chapter 5](#).

4.1 Strategy and Metrics

4.1.1 Dataset Statistics

The SVI manifest described in [section 3.2.2](#) contains 10,104 buildings and 47,238 images. After linkage with the quality-controlled and TABULA-NL-matched building-level reference table, 88 images associated with 18 buildings are excluded from model development and evaluation because of invalid 3DBAG geometry or unsuccessful archetype matching. All experiments in this chapter therefore use the final population of **10,086 buildings and 47,150 images**, comprising 8,068 buildings in the development set and 2,018 buildings in the frozen hold-out set. The values 10,104 and 10,086 do not represent contradictory counts of the same set: the former is the SVI data-delivery scope after image curation, whereas the latter is the modelling sample with complete experimental inputs and reference targets.

Table 4.1: City composition and fixed data partition of the experimental dataset

City	Dev. buildings	Dev. images	Hold-out buildings	Hold-out images	Total buildings	Total images
Amsterdam	6,416	29,309	1,595	7,215	8,011	36,524
Rotterdam	1,129	5,864	286	1,460	1,415	7,324
Utrecht	384	1,778	103	454	487	2,232
Delft	139	871	34	199	173	1,070
Total	8,068	37,822	2,018	9,328	10,086	47,150

The experimental sample is highly concentrated geographically: Amsterdam contains 8,011 buildings, corresponding to 79.4% of the experimental dataset. Its building-type composition also differs substantially from that of the eligible four-city registry reference pool, which comprises 124,784 residential buildings with EPCs that permit TABULA-NL

matching. Apartment Blocks (AB) account for approximately 88% of the experimental sample, compared with approximately 54% of the reference pool. The EPC-label distributions of the two sets also differ, but this difference cannot be characterised simply as a bias towards better ratings in the experimental sample. Under the binary grouping used in this study, A–C accounts for 70.3% of the experimental sample and 76.5% of the reference pool. Subsequent results should therefore be interpreted as estimates for this SVI-covered sample rather than unbiased estimates for the residential building stock of the four cities.

This mismatch in sample composition is consistent with the contributor-driven, non-random spatial coverage of Mapillary and is also influenced by the study area, residential-eligibility criteria, and image-quality screening. The observed bias is not further attributed to specific contributor behaviour. The effect of pool composition on model performance is measured in [section 4.3.5](#), and the resulting limitation on the scope of the study is discussed in [section 5.2](#).

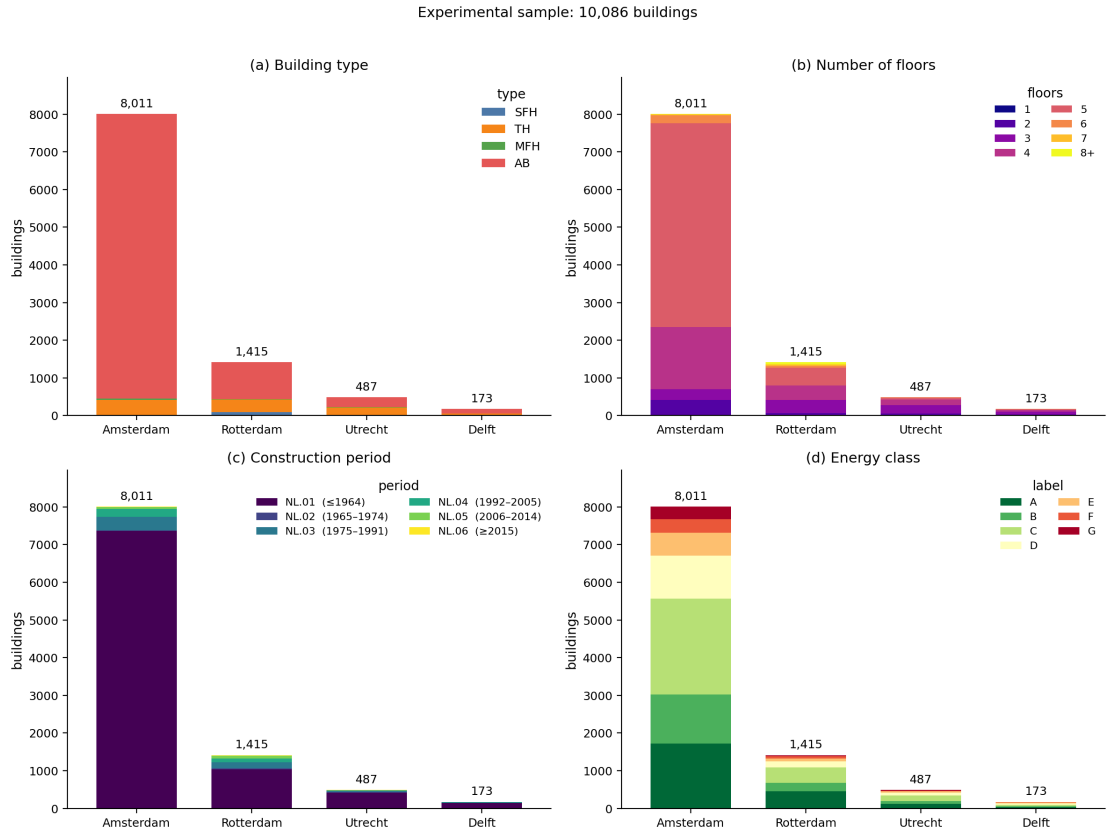


Figure 4.1: Experimental-sample attribute and EPC-label distributions by city.

4.1.2 Route Families and Evaluation Settings

Route families. Four evaluation routes distinguish a no-information baseline, reference-attribute inference, direct prediction from imagery, and attribute-mediated inference. The M identifiers denote inference routes rather than different datasets or downstream-model families.

Route	Input attributes	Inference procedure	Role in the comparison
M0	No building attributes or image input	Output the majority class of the corresponding development training data for all evaluation buildings	No-information baseline
M1	Reference building type, construction year, floor count, four U-values obtained from the reference TABULA-NL cell, and city	Input the eight reference-derived features to the fixed LightGBM EPC classifier	Reference-attribute baseline
M2	Street-view images	Predict the binary or seven-class EPC label directly from imagery without attribute prediction, TABULA lookup, or LightGBM	End-to-end comparator
M3	Vision-predicted building type, construction year, floor count, four U-values obtained from the predicted TABULA-NL cell, and city	Input the eight vision-derived features to the same fixed LightGBM EPC classifier used for M1	Attribute-mediated route

Evaluation settings (WIP). The primary evaluation uses the fixed building-level split defined in [section 3.2.3](#): 8,068 development buildings and a frozen hold-out set of 2,018 buildings. The five development folds support model selection and OOF analyses, whereas all pooled headline results are reported on the untouched hold-out set. A separate Amsterdam-held-out transfer test trains and selects all trainable components using 2,075 buildings from Rotterdam, Utrecht, and Delft and evaluates them on 8,011 Amsterdam buildings. The original pooled split is not retained in this setting. Because Amsterdam is the only held-out city, this experiment identifies a city-shift boundary rather than estimating average cross-city generalisation.

WIP summary. The ResNet-50, DINOv2, and InternVL3-2B configurations retain the parameter-update scopes, image counts, and multi-image aggregation procedures defined in [section 3.3](#) and [table 3.3](#). Each configuration produces an M3 attribute-mediated variant and a separately configured M2 direct-EPC comparator.

4.1.3 Prediction Tasks and Metrics (WIP)

WIP summary. Upstream evaluation covers building-type accuracy and macro-F1, construction-year MAE and TABULA-period accuracy, and floor-count MAE. Archetype assignment is assessed using joint-cell accuracy, a majority-cell baseline, and macro-cell recall. Binary A–C/D–G classification is the primary downstream task and uses ROC-AUC as its principal threshold-independent metric, with macro-F1 reported at the default and rate-matched operating points. Seven-class A–G prediction is retained as a diagnostic task using macro-F1, quadratic weighted kappa, exact and ± 1 accuracy, and class-distance MAE. Error attribution separates cases in which upstream attributes are correct but downstream classification fails from cases involving upstream attribute errors.

4.1.4 Physical-Consequence Metric (WIP)

WIP summary. The planned physical-consequence analysis reweights archetype-cell errors through the transmission heat-transfer coefficient H'_{tr} , calculated from predicted TABULA U-values and reconstructed 3DBAG envelope areas. It will report building-level error and floor-area-weighted stock-level heat-loss bias under a window-to-wall-ratio sensitivity analysis. This metric compares predicted and reference archetype cells; it is not a validation against measured energy consumption.

4.2 Quantitative Results

4.2.1 Upstream Attribute Extraction

[Table 4.2](#) compares the three vision configurations on the frozen hold-out set. Building type is evaluated using accuracy and macro-F1, construction year using MAE and TABULA-NL period accuracy, and floor count using MAE.

Table 4.2: Upstream attribute extraction on the hold-out set ($n = 2,018$)

Vision configuration	Type accuracy	Type macro-F1	Year MAE (years)	Period accuracy	Floor MAE
DINOv2, frozen backbone	0.901	0.520	9.45	0.901	0.377
ResNet-50, full fine-tuning	0.915	0.492	11.82	0.892	0.428
InternVL3-2B, zero-shot	0.495	0.246	30.65	0.769	0.708

ResNet-50 achieved the highest building-type accuracy, whereas DINOv2 obtained the highest type macro-F1 and the lowest errors for construction year and floor count. The difference between accuracy and macro-F1 indicates that aggregate type accuracy was strongly influenced by the dominant Apartment Block class; macro-F1 provides the more informative assessment of performance across the four building types. DINOv2 therefore provided the strongest overall balance across the three extracted attributes, although ResNet-50 retained a small advantage in type accuracy. InternVL3-2B performed substantially below both trained configurations on every reported metric. This result characterises the specific open-source 2B zero-shot configuration evaluated here and should not be interpreted as a general limitation of all zero-shot vision-language models.

4.2.2 Joint-Cell Assignment

DINOv2 and ResNet-50 achieved nearly identical joint-cell accuracies of 0.825 and 0.826, respectively, only approximately 3.5 percentage points above the majority-cell baseline of 0.790. DINOv2 obtained the higher macro-cell recall, but its performance remained strongly concentrated in the dominant cells: recall was 0.932 for AB | NL.01 and 0.789 for TH | NL.01, compared with 0.247 for AB | NL.03 and 0.074 for AB | NL.04. InternVL3-2B did not exceed the majority-cell baseline. These results show that high aggregate type and period accuracy did not translate into broad coverage of the TABULA-NL matrix; the trained configurations primarily provided reliable assignment for the cells most frequently represented in the hold-out set.

Table 4.3: Joint-cell assignment on the hold-out set (majority-cell baseline = 0.790)

Paradigm	Joint accuracy	Macro-cell recall
DINOv2 (frozen)	0.825	0.180
ResNet-50 (fine-tuned)	0.826	0.128
InternVL3-2B (zero-shot)	0.342	0.168

Figure placeholder

Per-cell recall heatmap for the 24 TABULA-NL archetype cells

Figure 4.2: Per-cell recall heatmap for the 24 TABULA-NL archetype cells

4.2.3 Downstream Route Comparison

The downstream routes were compared on the common frozen hold-out subset of $n = 2,018$ buildings for which all required predictions were available. The binary A–C versus D–G task is reported first as the primary downstream task. The original seven-class task is subsequently retained as a diagnostic analysis of fine-grained EPC prediction.

Binary Primary Task

[Table 4.4](#) compares the routes on the independently trained binary task, where D–G is treated as the positive class. Because the positive-class prevalence in the development set was approximately 0.297, results are reported at both the default 0.5 threshold and the rate-matched operating point defined in [section 4.1.3](#). ROC-AUC is used as the primary threshold-independent measure of discrimination.

Table 4.4: Binary downstream route comparison on the common hold-out set ($n = 2,018$)

Route	Inference configuration	Macro-F1 @ 0.5	Macro-F1 @ rate	ROC-AUC	PR-AUC
M0	Development-majority baseline	0.412	0.412	0.500	0.298
M1	Reference attributes	0.492	0.576	0.647	0.417
M2–DINOv2	Direct image-to-EPC, trained head	0.597	0.603	0.661	0.431
M2–ResNet-50	Direct image-to-EPC, full fine-tuning	0.566	0.565	0.615	0.403
M2–InternVL3	Direct zero-shot binary prompting	0.230	—	—	—
M3–DINOv2	Attribute-mediated, trained head	0.490	0.542	0.582	0.354
M3–ResNet-50	Attribute-mediated, full fine-tuning	0.488	0.528	0.570	0.346
M3–InternVL3	Attribute-mediated, zero-shot	0.528	0.502	0.535	0.322

WIP summary. The principal substitution comparison is M1 versus M3–DINOv2: ROC-AUC decreases from 0.647 to 0.582, corresponding to a substitution cost of 0.065. M2–DINOv2 achieves the strongest binary discrimination, with ROC-AUC 0.661, although part of its macro-F1 advantage depends on the operating point. The direct M2–InternVL3 prompt fails to provide class discrimination, whereas its attribute-mediated counterpart performs better; this finding remains specific to the evaluated model and prompt configuration.

Seven-Class Diagnostic Task

The original seven-class EPC task was retained to examine whether route differences persisted under the more fine-grained and label-imbalanced formulation. These results are treated as diagnostic rather than as the primary measure of practical screening performance.

Table 4.5: Seven-class performance and ordinal error structure on the common hold-out set ($n = 2,016$)

Route	Macro-F1	Exact accuracy	± 1 accuracy	Class-distance MAE	Quadratic κ
M0	0.068	0.310	0.612	1.198	0.000
M1	0.172	0.356	0.655	1.181	0.233
M2-DINOV2	0.213	0.266	0.544	1.625	0.236
M2-ResNet-50	0.201	0.263	0.570	1.537	0.188
M2-InternVL3	0.015	0.046	0.167	3.091	0.006
M3-DINOV2	0.150	0.292	0.589	1.373	0.083
M3-ResNet-50	0.149	0.282	0.587	1.381	0.091
M3-InternVL3	0.137	0.278	0.591	1.369	0.041

WIP summary. M2-DINOV2 records the highest macro-F1 and quadratic kappa, while M1 retains the strongest exact and ± 1 accuracy and the lowest class-distance MAE. M3-DINOV2 remains 0.022 below M1 in macro-F1 and shows a clearer deterioration in ordinal accuracy. The high ± 1 accuracy of M0 demonstrates that this metric is strongly affected by label imbalance and should be interpreted only as an error-structure diagnostic. Direct M2-InternVL3 prediction remains the principal failure case.

4.2.4 Physical-Consequence Scoring (WIP)

WIP summary. This section will compare the paradigms using building-level H'_{tr} error and stock-level heat-loss bias. The current draft indicates that the two trained configurations produce substantially lower physical error than the zero-shot configuration, while all three tend to overestimate aggregate heat loss. These calculations are independent of the EPC labels and therefore provide a separate assessment of archetype-substitution consequences.

Table 4.6: Physical-consequence metric on the hold-out set (WIP; WWR = 0.25; ranking unchanged across 0.15/0.25/0.35)

Paradigm	H'_{tr} MAE (W/(K·m ²))	95% CI	MAPE	Stock-level total heat-loss bias
DINOV2 (frozen)	0.150	[0.133, 0.169]	12.9%	+10.0%
ResNet-50 (fine-tuned)	0.157	[0.139, 0.177]	13.7%	+10.5%
InternVL3-2B (zero-shot)	0.474	[0.449, 0.500]	26.3%	+16.6%

4.2.5 Amsterdam-Held-Out Transfer Evaluation (WIP)

WIP summary. This transfer test trains all trainable components on 2,075 buildings from Rotterdam, Utrecht, and Delft and evaluates them on 8,011 Amsterdam buildings. The draft analysis indicates that building-type prediction transfers more reliably than construction-year prediction, joint-cell performance does not exceed Amsterdam’s majority-cell baseline, and the pooled advantage of direct image-to-EPC prediction disappears under city shift. Interpretation will separate visual-transfer failure from the limited discriminative information available in Amsterdam’s highly homogeneous target pool.

Table 4.7: Upstream degradation from pooled evaluation to LOCO (WIP; frozen DINOv2)

Metric	Pooled	LOCO	Δ
Type accuracy	0.901	0.897	-0.005
Year MAE (years)	9.45	19.50	$\times 2.1$
Period accuracy	0.901	0.876	-0.025
Joint accuracy	0.825	0.799	-0.026
			(majority-cell baseline = 0.868)

Table 4.8: LOCO downstream route comparison (WIP; test set = Amsterdam, $n = 8,011$)

Route	Seven-class macro-F1	Seven-class κ	Binary macro-F1 @ 0.5	Binary ROC-AUC
M0	0.050	0.000	0.410	0.500
M1 (reference attributes)	0.143	0.026	0.496	0.528
M2–DINOv2 (end-to-end)	0.140	0.050	0.499	0.528
M3–DINOv2 (decomposed)	0.141	0.030	0.490	0.505
M3–ResNet-50 (decomposed)	0.146	0.036	0.491	0.515

4.3 Qualitative and Diagnostic Analysis (WIP)

4.3.1 Error Mechanism: Regression to the Mean in Construction Year (WIP)

WIP summary. The planned diagnostic links construction-year regression towards the old-building majority to period-boundary errors, weak performance in minority archetype cells, overestimation of stock heat loss, and degradation under cross-city transfer.

4.3.2 Feature Ablation and Marginal Utility (WIP)

WIP summary. The draft leave-one-out analysis identifies construction year as the most influential downstream attribute. TABULA U-values add little conditional information once type and period are already included because they are deterministic functions of those attributes; this does not imply that physical U-values are intrinsically irrelevant.

Table 4.9: Attribute leave-one-out analysis and cumulative clean geometry (WIP; five-fold development OOF; baseline S_{full} macro-F1 = 0.181)

Operation	Δ macro-F1 (seven-class)	Δ macro-F1 (binary)
Remove construction year	-0.032	-0.030
Remove floor count	-0.012	-0.011
Remove building type	-0.003	+0.003
Remove U-values	-0.000 (conditionally redundant)	+0.002
Remove city	-0.009	-0.006
Add shape factor (3DBAG)	+0.027	—
Add floor area (3DBAG estimate)	+0.028	—
Add all four clean geometry features	+0.037 $\rightarrow$ macro-F1 $\approx$ 0.217 / $\kappa \approx$ 0.240	—

4.3.3 Extractability–Utility Audit (WIP)

WIP summary. This audit will compare how accurately candidate geometric features can be extracted from SVI with how much they improve downstream EPC prediction. The working interpretation is that geometry should be obtained from native 3D open-data sources when available, while SVI should focus on semantic attributes.

Table 4.10: Extractability $\times$ utility (WIP; L1: frozen-embedding probe OOF R^2 ; L2: ground-truth injection Δ F1)

Feature	L1 extractability R^2	L2 downstream utility Δ F1
Height	0.741	+0.023
Shared-wall ratio	0.460	+0.027
Volume	0.428	+0.020
Shape factor (clean version)	WIP: to be added	+0.027

4.3.4 Label-Quality Audit (WIP)

WIP summary. The planned audit examines the mismatch between dwelling-unit EPC issuance and building-level prediction. Multiple, internally inconsistent certificates for the same building impose an estimated oracle ceiling on building-level labels and require explicit reporting of the certificate-aggregation rule.

Table 4.11: Label-quality audit (WIP; manifest buildings and all raw EP-Online certificates)

Measure	Value
Share of <code>pand_id</code> values with multiple certificates	58.4%
Share with inconsistent labels among multi-certificate buildings	77%
Median modal share	0.50
Oracle ceiling (modal / latest)	0.79 / 0.88
Median within-building standard deviation of the energy indicator	35.8 kWh/(m ² ·yr)

4.3.5 Pool-Composition Audit (WIP)

WIP summary. This analysis will separate limitations caused by feature information from those caused by the apartment-dominated sample composition. The current draft indicates that changing the composition of the evaluation pool has a greater effect than increasing the sample size while retaining the same composition.

Table 4.12: Decomposition of pool-composition interventions (WIP; fixed feature set and hyperparameters; development baseline macro-F1 = 0.170, κ = 0.184, accuracy = 0.345)

Intervention	Δ macro-F1
Class weighting	+0.018
Break dominance of the majority cell (cell cap)	+0.064
Balance labels (class cap)	+0.039
Replace with the full-stock pool at the same sample size	+0.039
Increase data volume $\times 15$ while holding composition fixed	-0.001

Full-stock results for all 124,784 buildings: macro-F1 = 0.208, κ = 0.376, and accuracy = 0.440. **WIP.**

Chapter 5

Discussion and Conclusion (WIP)

This chapter will interpret the results reported in [chapter 4](#) without introducing new evidence. It will address the ceiling of seven-class EPC prediction, the unit-of-observation problem in EPC labels, the appropriate role of SVI in multimodal building-data enrichment, the study limitations, future work, and the final bounded conclusion.

5.1 Observations (WIP)

5.1.1 Seven-Class Ceiling: Task Structure Rather Than Visual Failure (WIP)

WIP summary. The proposed interpretation attributes the low seven-class ceiling to four interacting constraints: limited information in the available attributes, biased pool composition, unobserved building-system information, and inconsistent building-level EPC labels. The small gap between the reference-attribute and attribute-mediated routes suggests that visual substitution is not the sole cause of limited fine-grained performance.

5.1.2 Unit-of-Observation Mismatch in EPC Labels (WIP)

WIP summary. EPCs may be issued at dwelling-unit level while the model predicts at building level. The discussion will interpret within-building certificate disagreement as a failure of the assumed building-level homogeneity and recommend either unit-level prediction or an explicitly defined and sensitivity-tested building-level aggregation rule.

5.1.3 Boundary of the Role of SVI: Division of Labour Between Modalities (WIP)

WIP summary. The working principle is that SVI should supply semantic attributes, whereas reconstructed 3D open data should supply geometry. Attribute-mediated VLM inference appears more effective than direct EPC prompting for the evaluated InternVL3-2B configuration, but this result is bounded to the tested model and does not establish a universal rule for all VLMs.

5.2 Limitations (WIP)

WIP summary. The principal limitations are contributor-driven image-coverage bias, imperfect building-level EPC labels, evaluation of cross-city transfer using only Amsterdam as the target city, validation in a single country, operational simplifications in the TABULA mapping, incomplete coverage of model paradigms, and use of H'_{tr} as an internal physical reweighting rather than validation against measured energy consumption.

5.3 Future Work (WIP)

WIP summary. Priorities include testing image-based renovation cues through residual proxy labels, repeating leave-one-city-out evaluation for the remaining cities, collecting targeted suburban imagery, developing cross-national archetype mappings, and moving towards apartment-floor or dwelling-unit prediction.

5.4 Conclusion

This study evaluated whether SVI-derived building attributes can substitute for reference attributes in TABULA-based building-stock enrichment. A value-traceable semi-synthetic dataset linked attribute extraction, archetype assignment, physical parameters, and downstream EPC classification within a common comparison framework. The results provide clear but scope-bounded answers to the three research objectives.

For Objective 1, SVI supported the extraction of building type, construction year, and floor count and therefore served as an observational source for missing semantic building information. The frozen DINOv2 backbone provided the most balanced performance across the three attributes, ResNet-50 had a slight advantage only in building-type accuracy, and zero-shot InternVL3-2B was substantially weaker. These findings show that frozen self-supervised visual representations can provide effective features for multi-attribute building-profile enrichment.

For Objective 2, the extracted attributes had task-specific rather than uniform downstream utility. Construction year contributed the most information to EPC classification, while floor count made a smaller but directionally consistent contribution. Building type was structurally necessary for TABULA cell assignment but provided limited additional EPC discrimination in the apartment-dominated sample. Joint-cell evaluation further showed that strong individual-attribute performance primarily translated into reliable assignment of dominant archetype cells rather than balanced performance across the full TABULA matrix. The analysis therefore identified not only which attributes could be extracted, but

also which attributes should be prioritised and which downstream tasks benefited from them.

For Objective 3, SVI-derived attributes provided an informative partial substitute when reference metadata were missing, but they were not equivalent to reference attributes. The strongest attribute-mediated route retained discrimination above the no-information baseline in binary EPC classification, with a ROC-AUC substitution cost of 0.065 relative to the reference-attribute route. The physical-consequence analysis likewise showed that the trained configurations controlled archetype-derived heat-loss error more effectively than the zero-shot configuration. Cross-city evaluation, however, showed that substitution performance depended on alignment between the training and deployment distributions and may require target-domain calibration or more representative training coverage. This transfer boundary defines the conditions under which the approach applies without negating the information supplied by SVI under distribution-matched conditions.

Overall, SVI-derived attributes constitute a conditional enrichment source for buildings with missing reference metadata. Construction year provided the clearest downstream utility, while the frozen self-supervised representation offered the best overall balance between upstream extraction and control of physical consequences. The main contribution is a traceable evidence chain that separately quantifies which attributes are useful, the cost of substituting their source, and the conditions under which the substitution remains applicable. Within the Netherlands sample, which is dominated by Amsterdam apartment blocks, SVI-derived attributes retained a meaningful share of downstream information and supported interpretable TABULA-based building-stock enrichment.

Appendix A

Data-Integrity Audits

This appendix records the systematic audit of data-layer assumptions conducted before the analysis was finalised: what was checked, what was found, and how each issue was addressed. The main text presents only the corrected system and numerical results. This record retains the revision trail to support reproducibility and reconciliation with figures reported in earlier project-stage documents, including the interim presentation.

Table A.1: Audit items and actions

Audit item	Finding	Action
Feature provenance	“Compactness” and “floor area” were in fact EPC-certificate fields used to calculate the label. Their use as features constituted leakage, and the fields would be unavailable for buildings without certificates.	Removed the fields, recalculated them from reconstructed 3DBAG values, and updated all affected results.
TABULA lookup	The manually transcribed table contained a systematic thermal-resistance bias relative to the original workbook.	Replaced manual transcription with automated generation and value-level provenance. Classification results were unaffected because the lookup values are deterministic functions of $\text{type} \times \text{period}$; physical quantities were recalculated using the corrected values.
Label definition	The official ratings are binned according to the <code>forfaitair</code> variant of the indicator.	Aligned the regression target field. WIP: affected results are being recalculated.
Count reconciliation	Building counts reported at different stages referred to different sets.	Standardised the convention of reporting 10,104 buildings in the delivered SVI dataset and 10,086 buildings in the experimental dataset.
Certificate chronology	The database contains only current certificates, and only 0.1% of dwellings nationwide have a certificate time series.	Longitudinal renovation labels are infeasible; renovation detection is retained as future work in section 5.3 .
Nature of the geometry source	3DBAG contains automatically reconstructed outputs rather than field measurements. Floor count and the separation of internal and external walls are derived fields introduced after Peters et al. (2022).	Standardised the manuscript terminology as “reconstructed geometry”, documented the derivation of each field in table A.2 , and noted the need to report the version used.

The feature-provenance finding has been converted into the general eligibility rule stated in [section 3.5](#): the provenance of every candidate feature must be verified in advance, and any field used to calculate the target label must be excluded.

Table A.2: Definitions of derived 3DBAG fields (corresponding to the source-characterisation note in [section 3.2.1](#))

Field used in this study	Definition	Note
building_height	$b3_h_max - b3_h_maaiveld$	Ground elevation is the fifth percentile of ground-point elevations.
num_floors_estimated	$b3_bouwlagen$; if missing, $\text{round}(\text{building_height} / 3.0)$	Assumes a floor-to-floor height of 3.0 m.
floor_area_estimated	$b3_opp_grond \times \text{num_floors_estimated}$	Replaces the EPC-certificate field used incorrectly in the earlier analysis.
shared_ratio	$b3_opp_scheidingsmuur / (b3_opp_buitenmuur + b3_opp_scheidingsmuur)$	Classification statistic derived from reconstructed surfaces.
shape_factor	$\text{envelope_area} / \text{volume}$	A comparison variant including shared walls is also retained.

WIP: Add the identifier of the 3DBAG version used.

Appendix B

Supplementary Label-Collapse Comparisons

The results in this appendix apply post-hoc transformations to the seven-class predictions. They are not equivalent to the independently trained binary models reported in [table 4.4](#) and are therefore excluded from the primary downstream route comparison.

Table B.1: Literature-aligned label-collapse results

Route	Sun split macro-F1	Mayer split macro-F1
M0	0.412	0.457
M1	0.514	0.502
M2-DINOv2	0.583	0.562
M3-DINOv2	0.498	0.488

The Sun split uses A–C versus D–G, whereas the Mayer split uses A–D versus E–G. Because these values are obtained by collapsing seven-class predictions rather than retraining a binary classifier, they should not be compared directly with the binary results in [table 4.4](#).

References

- Algemene Rekenkamer. (2016). *Energie label voor koopwoningen*. Den Haag: Algemene Rekenkamer.
- Buckley, N., Mills, G., Reinhart, C., & Berzolla, Z. M. (2021). Using urban building energy modelling (UBEM) to support the new European Union's Green Deal: Case study of Dublin Ireland. *Energy and Buildings*, 247, 111115. <https://doi.org/10.1016/j.enbuild.2021.111115>
- CA EPBD. (n.d.). *Netherlands country page*. Retrieved 2026, from <https://www.ca-epbd.eu/countries/netherlands>
- CBS (Statistics Netherlands). (n.d.). *Kerncijfers wijken en buurten* [Key figures districts and neighbourhoods]. <https://www.cbs.nl/nl-nl/dossier/nederland-regionaal/wijk-en-buurtstatistieken>
- Chen, Z., Wang, W., Cao, Y., Liu, Y., Gao, Z., Cui, E., Zhu, J., Ye, S., Tian, H., Liu, Z., et al. (2024). Expanding performance boundaries of open-source multimodal models with model, data, and test-time scaling. *arXiv:2412.05271*.
- Concerted Action EPBD. (2020). *Implementation of the EPBD in The Netherlands — Status in 2020* (G. van Cruchten, Author; on behalf of Netherlands Enterprise Agency, RVO). Concerted Action EPBD.
- Dabrock, K., Ulken, J., Pflugradt, N., Weinand, J. M., & Stolten, D. (2025). Generating a nationwide residential building types dataset using machine learning.
- Dalach, A., Wang, Z., Nousias, S., & Borrmann, A. (2025). Building Energy Graphs (BEGs): How Graph Representation Design Impacts GNN-Based Building Energy Prediction.
- Dukai, B., Peters, R., Vitalis, S., van Liempt, J., & Stoter, J. (2021). Quality assessment of a large dataset containing automatically reconstructed 3D building models. *The International Archives of the Photogrammetry, Remote Sensing and Spatial Information Sciences*, XLVI-4/W4-2021, 17–24.
- European Commission. (2024). *Energy Performance of Buildings Directive (Directive (EU) 2024/1275)*. Official Journal of the European Union. <http://data.europa.eu/eli/dir/2024/1275/oj>
- Hettinga, S., Van 't Veer, R., & Boter, J. (2023). Large scale energy labelling with models: The EU TABULA model versus machine learning with open data. *Energy*, 264, 126175. <https://doi.org/10.1016/j.energy.2022.126175>

Ke, G., Meng, Q., Finley, T., Wang, T., Chen, W., Ma, W., Ye, Q., & Liu, T.-Y. (2017). LightGBM: A highly efficient gradient boosting decision tree. *Advances in Neural Information Processing Systems*, 30.

Khayatian, F., Sarto, L., & Dall'O', G. (2016). Application of neural networks for evaluating energy performance certificates of residential buildings. *Energy and Buildings*, 125, 45–54. <https://doi.org/10.1016/j.enbuild.2016.04.067>

Liang, X., Xie, J., Zhao, T., Stouffs, R., & Biljecki, F. (2025). OpenFACADES: An open framework for architectural caption and attribute data enrichment via street view imagery.

Liu, M., Gou, Z., & Zhao, Q. (2025). Enhancing building performance evaluation through Google Street View: A multimodal transfer learning framework. *Energy and Buildings*, 344, 115968. <https://doi.org/10.1016/j.enbuild.2025.115968>

Loga, T., Stein, B., & Diefenbach, N. (2016). TABULA building typologies in 20 European countries — Making energy-related features of residential building stocks comparable. *Energy and Buildings*, 132, 4–12. <https://doi.org/10.1016/j.enbuild.2016.06.094>

Mayer, K., Haas, L., Huang, T., Bernabé-Moreno, J., Rajagopal, R., & Fischer, M. (2023). Estimating building energy efficiency from street view imagery, aerial imagery, and land surface temperature data. *Applied Energy*, 333, 120542. <https://doi.org/10.1016/j.apenergy.2022.120542>

Mayer, K., Heilborn, G., & Fischer, M. (2023). Classifying building energy efficiency from street view and aerial images in Denmark. *Proceedings of the 10th ACM International Conference on Systems for Energy-Efficient Buildings, Cities, and Transportation (BuildSys '23)*, 220–223. <https://doi.org/10.1145/3600100.3623723>

Mapillary. (2026). *Mapillary: Street-level imagery platform*. <https://www.mapillary.com> (imagery accessed May 2026; licensed CC BY-SA 4.0).

Oquab, M., Darcet, T., Moutakanni, T., Vo, H., Szafraniec, M., Khalidov, V., Fernandez, P., Haziza, D., Massa, F., El-Nouby, A., et al. (2023). DINOv2: Learning robust visual features without supervision. *arXiv:2304.07193*.

Pan, Y., Lu, L., Wang, M., Reja, V. K., Noichl, F., Borrmann, A., & Brilakis, I. (n.d.). Zero-shot learning with vision-language models for estimating building energy efficiency from street view images. ASCE proceedings paper. **WIP: publication year to be confirmed.**

Peters, R., Dukai, B., Vitalis, S., van Liempt, J., & Stoter, J. (2022). Automated 3D reconstruction of LoD2 and LoD1 models for all 10 million buildings of the Netherlands. *Photogrammetric Engineering & Remote Sensing*, 88(3), 165–170. <https://doi.org/10.14358/PERS.21-00032R2>

Reinhart, C. F., & Cerezo Davila, C. (2016). Urban building energy modeling — A review of a nascent field. *Building and Environment*, 97, 196–202. <https://doi.org/10.1016/j.buildenv.2015.12.001>

RVO (Rijksdienst voor Ondernemend Nederland). (2011). *Example Dwellings 2011: Existing Buildings (Voorbeeldwoningen 2011: Bestaande bouw)*. Utrecht: RVO.

Sun, M., Hou, C., Li, Q., Zhang, F., Bardhan, R., & Zhao, Q. (2026). Deciphering exterior: building energy efficiency prediction with emerging urban big data. *npj Urban Sustainability*, 6, 38. <https://doi.org/10.1038/s42949-026-00348-7>

TABULA Project Team. (2012). *Typology Approach for Building Stock Energy Assessment. Main Results of the TABULA Project — Final Project Report (Deliverable D2)*. Darmstadt: IWU. https://episcopes.eu/fileadmin/tabula/public/docs/report/TABULA_FinalReport.pdf

Wurm, M., Droin, A., Stark, T., Geiß, C., Sulzer, W., & Taubenböck, H. (2021). Deep learning-based generation of building stock data from remote sensing for urban heat demand modeling. *ISPRS International Journal of Geo-Information*, 10(1), 23. <https://doi.org/10.3390/ijgi10010023>

Zeng, Z., Goo, J. M., Wang, X., Chi, B., Wang, M., & Boehm, J. (2024). Zero-Shot Building Age Classification from Facade Image Using GPT-4. *The International Archives of the Photogrammetry, Remote Sensing and Spatial Information Sciences*, XLVIII-2-2024, 457–464. <https://doi.org/10.5194/isprs-archives-XLVIII-2-2024-457-2024>

Declaration

I hereby affirm that I have independently written the thesis submitted by me and have not used any sources or aids other than those indicated.

Location, Date, Signature